Question ID: 0121a235

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	p(x)
-2	5
-1	0
0	-3
1	-1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

Answer

A. $(x-3)$

B. $(x+3)$

C. $(x-1)(x+2)$

D. $(x+1)(x-2)$

Correct Answer: D

Rationale

Choice D is correct. According to the table, when x is -1 or 2, $p(x)=0$. Therefore, two x-intercepts of the graph of p are $(-1,0)$ and $(2,0)$. Since $(-1,0)$ and $(2,0)$ are x-intercepts, it follows that $(x+1)$ and $(x-2)$ are factors of the polynomial equation. This is because when $x=-1$, the value of $x+1$ is 0. Similarly, when $x=2$, the value of $x-2$ is 0. Therefore, the product $(x+1)(x-2)$ is a factor of the polynomial function p.

Choice A is incorrect. The factor $x-3$ corresponds to an x-intercept of $(3,0)$, which isn't present in the table. Choice B is incorrect. The factor $x+3$ corresponds to an x-intercept of $(-3,0)$, which isn't present in the table. Choice C is incorrect. The factors $x-1$ and $x+2$ correspond to x-intercepts $(1,0)$ and $(-2,0)$, respectively, which aren't present in the table.